

Multi-projection specs

Main goal

The provided code allows to:

- generate a gnomonic projection image according to : Fh, Fv, lon, lat, which mean respectively horizontal field of view, vertical field of view, longitude, and latitude.
- the resolution of the projection is set to 'target_mp', which defaults to 0.48 MP. The specific width Wp and height Hp is computed such that the projected image has the same aspect ratio of Fh and Fv. That is: $Fh/Fv = Wp/Hp$.

the new script named `multi-persp.py` will generate multiple projections instead of letting the user interact using slides. The details are defined in this document:

Inputs

The input to the script are:

- `proj_nbr` : an integer number of projections to do
 - must satisfy this criteria: either pair (e.g. 6, 8) or `sqrt(proj_nbr)` is an integer. Otherwise, through suitable error mentioning that the number `proj_nbr` will not allow generating a grid of composite image of multiple rows without having missing grids without images. you could adjust my expression to be more precise
- Field of view: `fov_h` and `fov_v` : common for all `proj_nbr` projections
- latitude: common for all `proj_nbr` projections
- longitude:
 - starting longitude `lon_0` , i.e. longitude of the first projection
 - step longitude `lon_step` , i.e., the longitude of the i-th projection is `lon_i = lon_k + lon_step` , where `k = i - 1`
- `grid` : a tuple specifying (rows, cols) of the composite image to generate. The first projection should be put on the top left corner
- `comp_sz` : destination size of the composite image. Default to `640×640` (yolov8 input size)

Outputs

- create a new folder (if uncreated) named: `multi-persp-out`.
 - Inside of this folder are subfolders named: `pconf-i`, where `i` is a counter of the folders within multi-persp-out prefixed with `pconf-`.
- the subfolder `pconf-i` stores results of a specific projection configuration, specified by the previously mentioned 'Inputs' section above. There results are:
 - a log file named `log-pconf-i` storing:
 - the input configuration used, e.g., number of projections, field of view, grid, etc.
 - the log file should also contain a configuration that was not specified by the user, such as:
 - how much intersection the used configuration leads to between two adjacent projections
 - how much field of view was not covered by the used configuration
 - you are welcome if you have any other useful suggestions in this context
 - all generated projections, each projection is stored in one png file, according to this format : `persp-i-j`, where `i` is already explained, and `j` is equal to `0, 1, ...,proj_nbr`
 - a composite image stacking all projections into the rows and cols specified in the `grid` input argument, with the first projection being on the top left corner. This image is stored in two versions:
 - original size obtained after stacking the projected images: `or-composit-i.png`
 - resized image `rz-composit-i.png`, which simply resizes the original composite image to destination size specified in `comp_sz`
 - fisheye image will be stored into two versions:
 - `fish-i.png` (original fisheye processed, no resize, just restore it here)
 - `fovs-fish-i.png` (original fisheye overlayed by the field of view of each projection). The current script draws one field of view in green. We need to have now `proj_nbr` field of views remapped on the fisheye image, where each fov corresponds to a clear distinct color.